

TARANG BIO-OCEANOGRAPHIC RESEARCH SUITE

Scientific Data Extraction, Sensor Synthesis & Causal Modeling Methodology

Target Audience: Fisheries Scientists, Oceanographers, Marine Resource Managers	Coverage: 2018–2024 (84 monthly observation cycles)
Primary Domains: Satellite Ocean Color, SST Climatology, CMFRI Landings	Geographic Scope: 5 Indian Maritime Sectors (Malabar, Mannar, Saurashtra, Konkan, Coromandel)

1. Executive Overview & Problem Statement

A central problem in coastal marine resource management is answering: *'Why has fish productivity declined in a specific coastal region?'* Standard fishing advisories provide only operational short-term forecasts (e.g. today's wave height or immediate potential fishing zones), leaving researchers without longitudinal ecological diagnostic capabilities. Tarang bridges this gap by cross-referencing multi-sensor satellite oceanography (thermal regimes and phytoplankton biomass) with long-term commercial landing statistics to automatically isolate causal mechanisms (e.g., Marine Heatwave thermal displacement, delayed upwelling, or recruitment overfishing).

2. Primary Data Sources & Scientific Specifications

Parameter	Agency & Sensor	Resolution & Cadence	Extraction Protocol / Endpoint	Analytical Role
Sea Surface Temperature (SST)	NOAA / NCEI & INCOIS (AVHRR + In-situ OISST v2.1)	0.25° grid (~25 km) Daily binned to Monthly	REST / OPeNDAP Griddap: coastwatch.pfeg.noaa.gov/erddap	Thermal regime, stratification, thermocline depth, Marine Heatwaves (MHW $\geq +1.5^{\circ}\text{C}$).
Chlorophyll-a (CHL)	ISRO / INCOIS Oceansat-2 OCM + NASA MODIS-Aqua / VIIRS	4 km & 9 km grids Monthly Level-3 composite	NASA OB.DAAC & INCOIS Ocean Color Portal (SMI arrays)	Primary productivity proxy, diatom bloom onset, upwelling efficiency, trophic deficits (Z ≤ -1.5).
Commercial Fish Landings & CPUE	ICAR - Central Marine Fisheries Research Institute (CMFRI)	Maritime State / Region Monthly aggregated (2018–2024)	CMFRI National Marine Fish Landings Bulletins & FIMS	Empirical harvest telemetry: metric tonnes, Catch Per Unit Effort (kg/hr), MSY baseline benchmarks.
Near Real-Time Marine Weather	ECMWF ERA5 + DWD ICON via Open-Meteo	0.1° grid (~11 km) Hourly operational forecast	Open-Meteo Marine & Forecast API (REST)	Significant wave height (Hs), swell period, surface wind speeds, real-time artisanal boat safety.

3. Step-by-Step Data Extraction & Processing Pipeline

Step 1: Spatial Sector Delimitation & Bathymetric Masking

Each of the 5 coastal regions is bounded by a dedicated maritime geofence polygon (e.g. Malabar Coast: 8.5°N–12.0°N, 75.0°E–77.0°E). A bathymetric mask (< 200 m continental shelf isobath) is applied to discard open-ocean pelagic pixels, ensuring observations reflect the shallow coastal shelf where artisanal and mechanized near-shore fisheries operate.

Step 2: SST Extraction & Climatological Anomaly Computation

Monthly Sea Surface Temperature is extracted via NOAA's OISST v2.1 reanalysis. For each month m , the observed SST is benchmarked against the 30-year climatological baseline $\mu_{\text{SST}, m}$ (1991–2020 average for that calendar month):

$$\Delta\text{SST}_m = \text{SST}_{\text{obs}, m} - \mu_{\text{baseline}, m}$$

A **Marine Heatwave (MHW)** is formally flagged when $\Delta\text{SST} \geq +1.5^{\circ}\text{C}$ persists over the region.

Step 3: Chlorophyll-a Normalization & Z-Score Anomaly Detection

Satellite ocean color chlorophyll-a concentrations ($\mu\text{g/L}$ or mg/m^3) exhibit high seasonal variability driven by the southwest monsoon upwelling. To identify genuine primary productivity deficits rather than normal winter troughs, observations are converted to standardized

monthly Z-scores:

$$Z_{CHL, m} = (CHL_{obs, m} - \mu_{CHL, m}) / \sigma_{CHL, m}$$

A **Trophic Deficit** is classified when $Z_{CHL} \leq -1.5$, signaling significant primary phytoplankton collapse.

Step 4: CMFRI Catch Statistics Alignment & CPUE Scaling

Monthly commercial landings are extracted from CMFRI national bulletins. Landings are correlated with effort hours to compute Catch Per Unit Effort (CPUE in kg/hour). Landings are benchmarked against historical 5-year rolling averages to identify stock decline windows (> 30% drop below replacement).

4. Mathematical Formulations & Causal Attribution Engine

The automated diagnosis engine in backend/tools/ecosystem_analytics.py computes statistical relationships in real-time across user-selected windows:

Statistical Metric	Mathematical Formula	Oceanographic / Ecological Interpretation
Trophic Coupling (CHL ↔ Catch)	$r_{CHL, Catch} = \frac{\sum (x - \mu_x)(y - \mu_y)}{\sqrt{\sum (x - \mu_x)^2} \sqrt{\sum (y - \mu_y)^2}}$	Quantifies bottom-up biological dependency. High positive correlation confirms that larval survival and schooling biomass are directly sustained by coastal diatom blooms.
Thermal Displacement (SST ↔ Catch)	$r_{\Delta SST, Catch}$ (Pearson correlation with p -value)	Negative correlation indicates thermal displacement: anomalous surface warming forces stenothermal pelagics (Oil Sardine) to dive below 30 m or migrate poleward, escaping artisanal gear.
Upwelling Efficiency (SST ↔ CHL)	$r_{SST, CHL}$ (Monthly lagged correlation)	Evaluates coastal upwelling strength. Strong negative correlation proves that cold, nutrient-rich deep water is the essential prerequisite for primary biological production.
Composite Ecological Stress Index	$Score = w_1(MHW_ratio) + w_2(CHL_deficit_ratio) + w_3(Catch_drop_ratio)$	Normalized 0–100 score classifying regional marine ecosystem health into Nominal, Moderate Stress, or Severe Disruption.

5. Tarang Dual-Track Data Architecture

Tarang employs a decoupled architecture separating high-speed longitudinal research analytics from real-time vessel safety operations:

Dimension	Track A: Researcher Longitudinal Engine	Track B: Operational Fishing Safety Engine
Primary User	Fisheries researchers, marine biologists, state fisheries depts.	Artisanal fishermen, boat masters, harbour safety officers.
Temporal Scope	2018–2024 (Monthly continuous longitudinal series).	Real-time (Nowcast + 24–72 hour operational forecast).
Storage & Access	Pre-compiled unified schema (fisheries_productivity_timeseries.json) with instant client-side SVG rendering.	Live REST APIs (INCOIS ERDDAP, Open-Meteo Marine, IMD, GDACS) with 60-minute in-memory TTL caching.
AI Interface	Dedicated Bio-Oceanographic AI Fellow grounded in empirical correlation vectors, MHW counts, and CMFRI reports.	Multilingual Voice & SMS Agent (Hindi, Tamil, Malayalam, Bengali) for daily wave & safety advisories.

6. Bio-Oceanographic AI Grounding & Anti-Hallucination Protocol

To ensure the Researcher AI Chatbot remains scientifically rigorous and never produces generic or fabricated answers, Tarang enforces a strict context injection protocol:

- Zero Hallucination Grounding:** The user's query is never sent in isolation. The system dynamically computes the exact Pearson r , p -values, MHW episode counts, and deficit months for the active region and time window, injecting them directly into the LLM system prompt.
- Causal Attribution Taxonomy:** The model is constrained to evaluate three scientifically validated marine decline mechanisms: (a) Thermal Stratification & Thermocline Deepening, (b) Trophic Larval Starvation via Diatom Bloom Delay, and (c) Post-Spawning Recruitment Overfishing.
- Policy & Intervention Linking:** Every diagnostic attribution maps to actionable management recommendations (e.g., dynamic seasonal ban adjustments, minimum legal size enforcement, and artificial reef nursery deployment).